De La Salle Holy Cross College

Grade 10 Mathematics

Mid-Year Examination - Paper 1 - July 2021

Examiners: Mathematics Department Total: 105

Moderators: Grade 10 Mathematics Teachers Time: 2 hours

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Instructions and Information:

- This assessment consists of 13 pages which include 8 questions.
- Answer every question in the spaces provided.
- You may use any permitted calculator. Ensure that it is set in degree mode.
- All necessary working-out must be shown clearly; providing only final answers does not necessarily guarantee full marks.
- Diagrams and graphs are not necessarily drawn to scale.
- Write neatly and legibly; anything that is unclear will not be marked.

	Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	Total
Mark	KR	PG	PG	LC	KR	LC	LC	LC	
Total	**25**	**19**	**12**	**12**	**9**	**8**	**6**	**14**	**105**

QUESTION 1 [25 Marks]

(a) Consider the following numbers:

$\sqrt{27}$ $\sqrt{-27}$ $\sqrt[3]{-27}$

Write down the number which is:

(1) irrational. (1)

$\sqrt{27}$ ✓a

(2) non-real. (1)

$\sqrt{-27}$ ✓a

(b) Simplify:

(1) $(x+3)(x-3)+(x-4)^2$ (4)

$x^2 - 9 + (x^2 - 8x + 16)$

$= 2x^2 - 8x + 7$ ✓ca ✓ca

(2) $\left(x-\frac{4}{x}\right)\left(x^2+4+\frac{16}{x^2}\right)$ (2)

$x^3 - \frac{64}{x^3}$ ✓ ✓

(3) $\frac{2x^2+x-6}{2x^2+4x}$ (3)

$= \frac{(2x-3)(x+2)}{2x(x+2)}$ ✓a ✓a

$= \frac{2x-3}{2x}$ ✓ca

(or $1 - \frac{3}{2x}$)

(4) $\frac{3}{m-4}+\frac{2}{m+3}-\frac{21}{m^2-m-12}$ (6)

$(m-4)(m+3)$

$$\frac{3(m+3)+2(m-4)-21}{(m-4)(m+3)}$$

$$=\frac{3m+9+2m-8-21}{(m-4)(m+3)}$$

$$=\frac{5m-20}{(m-4)(m+3)}=\frac{5(m-4)}{(m-4)(m+3)}$$

$$=\frac{5}{m+3}$$

(5) $\sqrt{4x^2-20x+25}$ (2)

$$=\sqrt{(2x-5)^2}$$

$$=2x-5$$

(6) $\frac{x^2-7x+12}{x^2-16}\div\frac{9-x^2}{4+x}$ (6)

$$\frac{(x-4)(x-3)}{(x-4)(x+4)}\times\frac{(4+x)}{(3-x)(3+x)}$$

$$=\frac{-1}{x+3}\quad\text{or}\quad\frac{1}{-x-3}$$

QUESTION 2 **[19 Marks]**

(a) Given the inequality $4 - 3x > -x + 7$

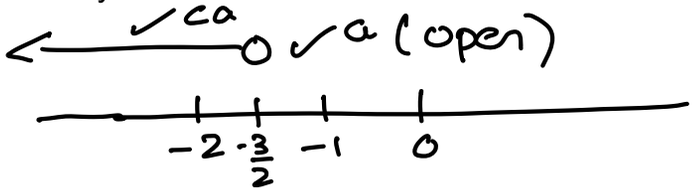
(1) Solve for x. (2)

$4 - 3x > -x + 7$

$-3 > 2x$ ✓

$-\frac{3}{2} > x$ ✓

(2) Represent your answer on a number line for $x \in R$. (2)



(3) What is the greatest integer represented by this inequality? (1)

$x = -2$ ✓ca

(b) Solve for m, $(m \neq 0)$:

(1) $\frac{m}{3} - \frac{m-2}{4} = 2 - \frac{m}{2}$ (4)

LCD ✓12

$4m - 3(m-2) = 24 - 6m$ ✓

$4m - 3m + 6 = 24 - 6m$

$7m = 18$

$m = \frac{18}{7}$

(2) $15m^2 - 2m = 1$ (4)

$15m^2 - 2m - 1 = 0$ ✓

$(5m + 1)(3m - 1) = 0$ ✓

$m = -\frac{1}{5}$ ✓ or $m = \frac{1}{3}$ ✓

(3) $5 \cdot 32^m = 40$ (3)

$32^m = 8$ ✓

$2^{5m} = 2^3$ pF ✓

$5m = 3$

$m = \frac{3}{5}$ ✓

(4) $k - \frac{p}{m} = t$ (3)

$k - t = \frac{p}{m}$ ✓ or $\frac{k-t}{p} = \frac{1}{m}$

$m = \frac{p}{k-t}$ ✓

QUESTION 3 [12 Marks]

Simplify: make sure your final answer has positive exponents.

(a) $\dfrac{125^{x} \times 25^{x+1}}{5^{5x+3}}$ (3)

$$\frac{5^{3x} \times 5^{2x+2}}{5^{5x+3}}$$

$$5^{-1} = \frac{1}{5}$$

(b) $\dfrac{a^{-3} \cdot b \cdot c^{4}}{2 \cdot b^{-5}} \div \dfrac{8(a \cdot b)^{3}}{(2 \cdot a^{-2} \cdot b^{3})^{4}}$ (6)

$$\frac{a^{-3} b c^{4}}{2 b^{-5}} \times \frac{16 a^{-8} b^{12}}{8 a^{3} b^{3}}$$

$$\frac{b^{15} c^{4}}{a^{14}}$$

(c) $\dfrac{3^{x+1} + 3^{x}}{27 \cdot 3^{-1+x}}$ (3)

$$= \frac{3^{x} \cdot 3 + 3^{x}}{27 \left(\frac{1}{3}\right) 3^{x}}$$ split up powers.

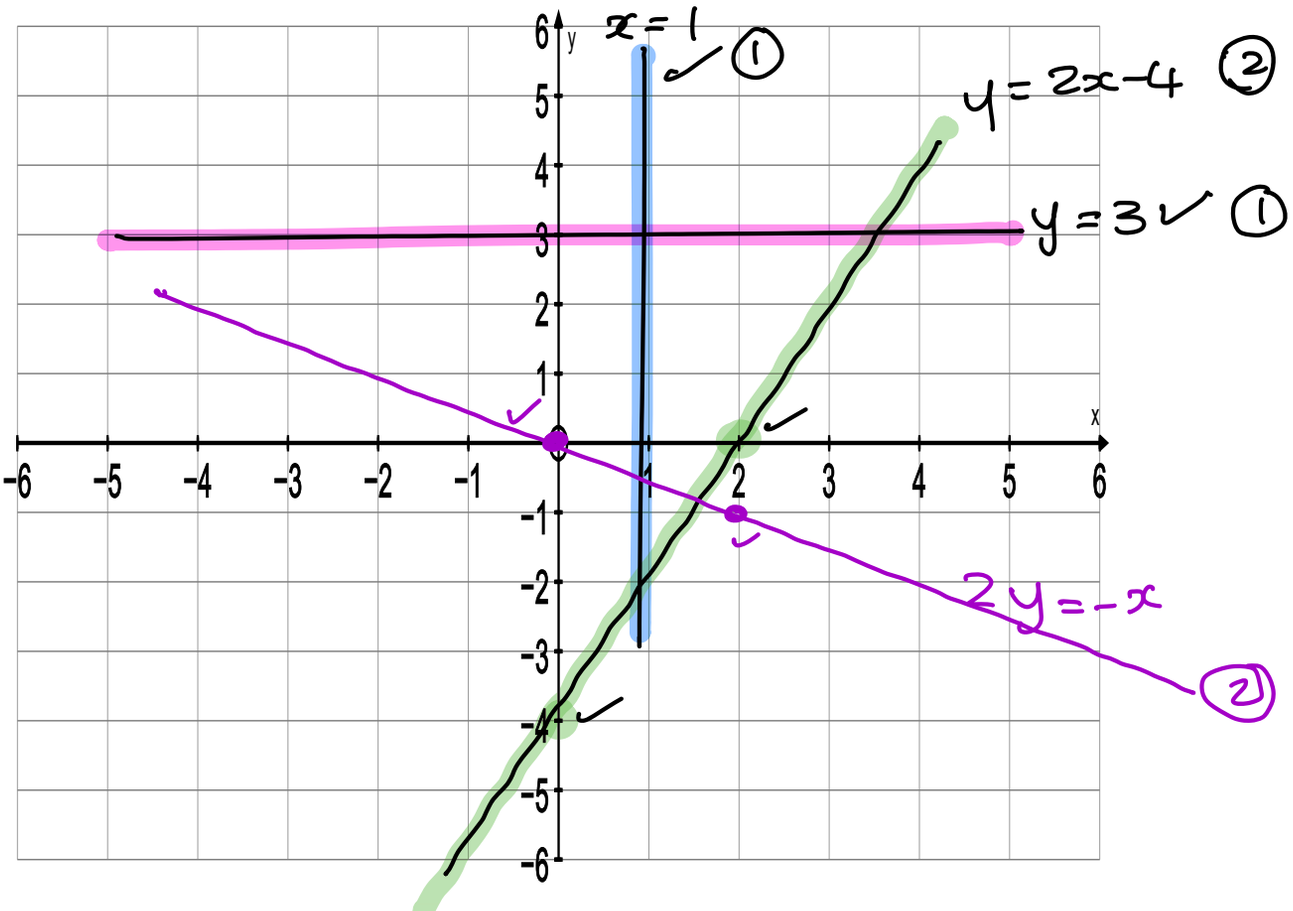
$$= \frac{3^{x} (4)}{9 \cdot 3^{x}}$$

$$= \frac{4}{9}$$

QUESTION 4 [12 Marks]

(a) On the set of axes below sketch the following straight line graphs. Label your graphs carefully. Please indicate the values of the intercepts with the axes carefully.

(1) $y = 3$
(2) $x = 1$
(3) $y = 2x - 4$
(4) $2y = -x$ (6)



(b) Write down the point of intersection of graphs (1) and (2) (2)

(1; 3)

(c) Determine algebraically (by calculation) the point of intersection of the lines $y = 2x - 4$ and $2y = -x$. (4)

$y = 2(-2y) - 4$
$y = -4y - 4$
$+5y = -4$
$y = \frac{-4}{5}$ ✓ $x = -2x - \frac{4}{5}$
$= \frac{8}{5}$
$\left(\frac{8}{5}; -\frac{4}{5}\right)$

$y = -\frac{1}{2}x$
$-\frac{1}{2}x = 2x - 4$
$-x = 4x - 8$
$-5x = -8$
$x = \frac{8}{5}$ ✓
$y = -\frac{1}{2} \times \frac{8}{5}$
$= -\frac{4}{5}$
$\left(\frac{8}{5}; -\frac{4}{5}\right)$

QUESTION 5 **[9 Marks]**

Factorise completely:

(a) $3x^4 + 30x^2 + 75$ (3)

a ✓ $3(x^4 + 10x^2 + 25)$ ✓

$3(x^2 + 5)(x^2 + 5)$ ✓

(b) $21x^2 - 22x - 8$ (2)

$(7x + 2)(3x - 4)$ a a

(c) $9a^2(x - y) + (y - x)$ (4)

$9a^2(x-y) + (x-y)$ ✓

$(x-y)(9a^2+1)$ ✓

$(x-y)(3a-1)(3a+1)$ ✓

QUESTION 6 [8 Marks]

The water level in a tank is recorded while the tank is being filled. The results have been recorded at five minute intervals, starting when the water in the tank is 3 cm deep:

	First reading	**After 5 min**	**After 10 min**	**After 15 min**	**After 20 min**
Level in *cm*	3	11	19	27	35

(a) What will the height of the water be after 25 minutes? (1)

43 cm ✓

(b) At what rate is the water level rising? Give your answer in cm/min. (2)

$\frac{99-3}{60} \overset{m}{=} \frac{96}{60} = \frac{8}{5}$ ✓ cm | min

(c) Determine a formula T_n that will describe the level in cm as shown by the values in the table above. (2)

$T_n = 8n$ ✓ $- 5$ ✓

(d) What will the height be after one hour? (1)

99 cm ✓ → $8 \times 13 - 5 = 99$

or 3, 11, 19, 27 99

(e) After how many minutes will the water level reach a total depth of 403 cm? (2)

$\frac{403-3}{8/5}$ ✓ ✓

= 250 minutes

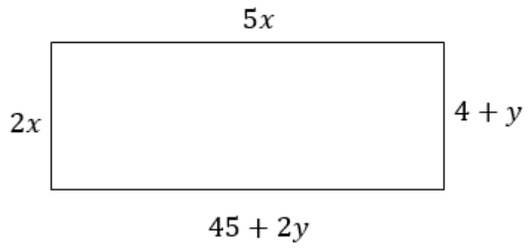
or $n = 51$ ✓ $(\frac{408}{8})$

$t = 5(n-1)$.

$\therefore t = 250$ ✓

QUESTION 7 [6 Marks]

Determine the values of x and y represented in the diagram of a rectangle drawn below.



(6)

$5x = 45 + 2y$

$5x = 45 + 2(2x - 4)$

$= 45 + 4x - 8$

$x = 37$

$2x = 4 + y$

$2x - 4 = y$

$y = 70$

QUESTION 8 [14 Marks]

(a) James was given the following calculation:

$$9785627^2 - 9785630 \times 9785624$$

His calculator gave him an answer of zero which is not correct.

Without doing long multiplication and without using your calculator explain to James how to find the correct answer. You must indicate the value of the correct answer. (6)

Let $x = 9785627$ ✓

$x^2 - (x+3)(x-3)$ ✓ ✓ ✓

$x^2 - x^2 + 9$ ✓

$= 9$ ✓

(b) Given $4^{x+y} = 64 \ and \ 3^{x-y-1} = 1$

(1) Show that $x + y = 3 \ and \ x - y = 1$ (4)

$4^{x+y} = 4^3$ ✓ $\quad 3^{x-y-1} = 3^0$ ✓

$x + y = 3$ $\quad x - y - 1 = 0$ ✓

$x - y = 1$

✓

(2) Hence determine the value of $3^x \cdot 5^y$. (4)

$2x = 4$

$x = 2$ ✓ $\quad \therefore y = 1$ ✓

$3^2 \times 5$ ✓

$= 45$ ✓ (4)

TOTAL: 105 marks

THIS PAGE IS BLANK AND MAY BE USED FOR ANY ADDITIONAL WORKING. PLEASE LABEL QUESTION NUMBERS CLEARLY.